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The impact of school activity space layout on children's physical activity levels during recess: An agent -based model computational approach ☆

Jiameng Cui [a,b], Xue Meng [a,b], Siming Qi [a,b], Junwen Fan [a,b], Wenxuan Yu [a,b], Hongxing Liu [a,b], Xiang Wang [a,b], Yu Zhang [a,b,*]

[a] School of Architecture and Design, Harbin Institute of Technology, Harbin,150001, PR China

[b] Key Laboratory of Cold Region Urban and Rural Human Settlement Environment Science and Technology, Ministry of Industry and Information Technology, Harbin 150001, PR China

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ABSTRACT

Children's physical activities are complex and influenced by their intricate environmental and social relationships. This complexity is particularly evident during school recess, where schools often struggle to facilitate moderate to vigorous physical activity among students. Therefore, this study aims to propose an Agent-Based Model (ABM) to investigate the levels, spatial distribution, and duration of physical activity during primary school recess.

Utilizing survey data from 3 Chinese primary schools (N = 801), this research established the parameters for the ABM. Subsequently, ABM was constructed based on a social force model to examine the impact of the evolution of social networks on children's physical activities during recess. Additionally, the study explored the impact of spatial environmental factors in the activity space on physical activity levels.

Simulation results indicate that the ABM more realistically simulated the situation of students' choices made for both activity space and physical activity during recess, highlighting the impact of spatial environmental factors on physical activities. The layout of activity spaces, particularly the distance and accessibility between the playground and school buildings, impacts both the intensity and duration of physical activity among students during recess. Furthermore, the study recommends promoting the design of activity spaces to encourage physical activity. The findings of this research provide specific evidence for understanding the factors contributing to children's physical activity during recess and investigating the impact of activity spaces on physical activity levels. Moreover, the results offer theoretical support for future research on models of children's physical activity.

1. Introduction

According to the World Health Organization (WHO), children and adolescents should do at least an average of 60 mins per day of moderate-to-vigorous-intensity (MVPA), mostly aerobic, physical activity, across the week. Global estimates indicate that in 2016, only 19 % of children aged 11–17 years achieved adequate levels of physical activity (PA) to meet the 2010 WHO recommendations, with trend data revealing minimal global improvement over the past decade [1]. PA levels were lowest in high-income Asia Pacific (11 % of boys and 4.4 % of girls) and highest among boys in Western countries (27.9 %) and girls in South Asia (22.5 %) [2]. This trend is confirmed by data from multiple countries, with figures showing 25 % in the USA [3], 18 % in England [4], and only 11 % in Scotland [5]. In China, the proportion of schoolchildren (aged 8–19 years) meeting the recommended MVPA increased modestly from 22 % in 2010 to 30 % in 2016 [6]. Chinese primary school students (aged 6–12 years) face challenges like low PA levels and obesity (34.1 % met MVPA guidelines in 2017; 24.2 % obesity rate in 2019). China has implemented health promotion initiatives in primary and secondary schools, emphasizing the reinforcement of physical education, health courses, and extracurricular exercise time.

Schools serve as primary venues for children's daily PA, with a significant portion occurring during recess [7,8]. Studies have confirmed that recess serves as a valuable opportunity for primary school students

* Corresponding author at: School of Architecture and Design, Harbin Institute of Technology, Harbin, 150001, PR China.
E-mail address: yu.zhang@hit.edu.cn (Y. Zhang).

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to increase their levels of PA [9]. Nearly 40 % of variability in primary school children's recess MVPA was explained by school environment and individual characteristics identified, such as grassed surfaces per child and shaded grassed surfaces [10]. Furthermore, variations in playground functionality, spatial layouts, topography, and vegetation design have been shown to significantly influence outdoor activity duration and PA levels across different grade levels [11]. Schools that are functionally diverse, well-vegetated, and equipped with various facilities, combined with rational spatial layouts, have been found to attract students to outdoor activities and promote higher levels of PA and social interactions [12].

In China, children typically attend school for about 8 h a day, 5 days a week, from 8:00 a.m. to 5:00 p.m. Given the increasing focus on academic performance, opportunities for children to engage in PA outside school hours are limited, making schools critical venues for promoting PA [13]. Most primary and secondary schools in China are located in high-density urban areas. These schools adopt an intensive approach in their campus designs, focusing on efficient land use and integration of building functions, which limits the layout of activity spaces. Due to physiological differences, such as body proportions and neuromuscular control systems, children exhibit shorter strides and limited activity ranges compared to adults [14]. The design of these spaces impacts children's ability to engage in diverse activities during their 10-minute recess, leading to a noticeable increase in sedentary behavior [15]. A study of 8316 Chinese students found that 64.24 % rarely went outdoors during recess [16].This, in turn, elevates the risk of overweight and obesity, with adverse consequences for children's health [6]. Addressing this issue and strategically designing school activity spaces to encourage PA are vital for improving children's overall well-being.

Agent-Based Modeling (ABM) has matured as an effective solution for space layout problem [17]. By simulating human actions and thoughts, ABM, with its goal-driven agents interacting autonomously with the environment, exhibits high accuracy in predicting collective behaviors [18,19]. ABM integrates a versatile simulation framework that includes space, time, and social networks [20,21]. It can simulate key PA measures such as type, spatial distribution, and duration, using agents to model individual behaviors. This paper focuses on utilizing ABM to simulate the spatial configurations of primary school activity spaces, specifically during recess, and examines how different layouts influence students' PA.

The study addresses two primary research aims: i) How to construct an ABM model for simulating students' PA during recess, including the establishment of the model's environment, agents, and decision logic? ii) What is the relationship between factors of activity spaces and primary school students' PA? What type of spatial layout design promotes increased PA levels?

This paper begins by reviewing the existing literature to understand the impact of PA on children's health, the factors impacting PA in school activity spaces, and the construction and application of ABM. Based on the findings from the literature, Section 2 presents the survey data and the analytical methods used to calibrate the agent, followed by the design, decision logic, and construction of the ABM model. Section 3 details the experiments conducted on the case study primary school and presents the experimental results. The study examines the impacts of different spatial layouts on the PA of primary school students. Finally, Section 4 discusses the findings, limitations, and potential directions for future research.

1.1. The impact of PA on children's health

American scholar Jane E. Clark's "Mountain of Motor Development" theory highlights the crucial period of 7 to 12 years for children's fundamental motor skills, representing a golden era for psychological and physical development [22]. Psychologically, Observations and statistical analyses by Slutzky and Simpkins reveal a positive correlation between increased time spent in team activities and children's improved self-perceived athletic abilities, positively affecting self-esteem [23]. LaForge-MacKenzie et al. collected data during the COVID-19 pandemic, indicating that children engaged in extracurricular activities or sports before the outbreak experienced fewer symptoms of anxiety, attention deficit, hyperactivity, and depression [24].

In general, longer periods of PA among children lead to greater health benefits, with particularly significant improvements for those with lower initial activity levels, as each additional minute of exercise yields disproportionately greater health gains [5]. Wang et al. randomized clinical trial showed that more time dedicated to extracurricular PA not only had no negative impact on primary students' math grades but also increased children's happiness index, physical fitness scores, and reduced vision problems [25]. Zahner et al. reported through random sampling and data analysis that children participating in PA scored significantly higher in all sports fitness indicators, enhancing aerobic fitness and reducing cardiovascular risk factors [26]. Basterfield et al.'s research demonstrated that regular childhood PA, including sports, correlated with increased bone density and skeletal muscle strength, providing long-term preventive effects on early adolescence obesity, manifesting results by the age of 12 years [27].

A study by Ridgers et al. found that recess activity can account for up to 40 % of the recommended 60 mins of daily MVPA [28]. Recess provides time for PA but can reduce academic class time, potentially impacting learning. However, research by Angela K. Dills shows that recess can improve classroom behavior and boost comprehension in younger students [29]. Since recess activity typically takes place outdoors, children may also benefit from increased vitamin D levels due to sunlight exposure [30], and lower blood pressure from contact with natural elements like greenery [31]. Additionally, more PA during recess may double the chances of meeting PA recommendations [32].

1.2. The impact of school activity space on PA

Horton and Reynolds defined activity space as the subset of locations individuals directly encounter in their daily activities within the action or cognitive space [33]. In primary school research, though there is no precise definition for activity space, it typically encompasses spaces like playgrounds, soccer fields, basketball courts, and play zones [34–36]. Some studies also consider areas such as corridors [37], courtyards [38], and in-between spaces, including small enclosures, edges and natural settings [39]. Considering the characteristics of Chinese primary schools, this study defines activity space as including corridors, courtyards, playgrounds, entrance square, and semi-indoor/outdoor platforms.

From a health promotion perspective, physical environmental factors are particularly important, as many can be easily modified through school design [40]. Sumiya and Nonaka demonstrated that modifying the spatial layout of play equipment on the same playground can boost children's PA levels [41]. Aminpour highlighted the role of school design in reducing conflicts among children engaged in PA, emphasizing that dedicated play zones for different age groups and sufficient buffer spaces can enhance student participation in PA [42]. Contemporary school designs in Australia, focusing on vertical schools that integrate learning and activity spaces through features like courtyards, stairs, and terraces, have proven to optimize child satisfaction by facilitating outdoor exposure and meeting diverse PA needs [43]. Henriette et al. compared PA levels during recess before and after campus renovations, finding that school layouts incorporating more functional and natural factors provided greater opportunities for PA among both boys and girls, potentially enhancing the utility and engagement of recess [44]. Studies on student movement patterns within schools show that activity routes are shaped by the spatial relationships between functional areas, such as zone placement, openings, and pathway [45].

Several studies have described the impact of schoolyard characteristics on PA and found various factors to be conducive to PA during recess, such as various play equipment [46,47], exposure to natural

[48], playground location and size [34,49]. Grunseit et al. found that increasing playground area to 25 m²s per student, along with suitable play equipment, significantly enhances PA among students in 43 primary schools in New South Wales, Australia [50]. Similarly, reducing playground density helps decrease sedentary time, particularly for less active children during recess [51]. Raney et al. identified several factors influencing PA during recess, including playground density, green space percentage, the number of unique play zones, types of games, area segregation, and shaded spaces, with their findings indicating that playgrounds featuring more distinct play zones, segregated areas, and shaded spaces promote MVPA [36].

In examining the impact of school environmental factors on children's PA during recess, most studies have relied on observational or self-reported measurements [52,53], with only a few utilizing objective methods such as video analysis, accelerometers, or GPS tracking [54,55]. This indicates a notable gap in methodologies for visualizing and modeling the spatial relationships between PA and the school environment. While these studies offer valuable insights into the influence of school spatial environments on PA from various perspectives, most research concentrates on playgrounds, often overlooking the issue of other activity space type, such as platform, courtyard and square. Furthermore, many studies primarily investigate environmental factors within the activity spaces themselves, neglecting the influence of the layout of these activity spaces within the school in relation to the teaching building on students' PA levels.

1.3. ABM research and applications in children's studies

In the 1960s, the exploration of complex systems spanned diverse fields, such as computer science, artificial intelligence, control theory, and chaos theory, laying the foundation for ABM [56]. ABM conceptualizes individuals as autonomous "agents" with distinct attributes and behavioral possibilities, engaging in adaptive interactions with their environment to generate collective behaviors at the group level [57,58]. ABM is extensively applied in building occupant models and social simulations [59]. Building occupant models simulate residents' behaviors to optimize architectural designs, simulate building performance, and predict human behavior under building control systems [17,60]. In social simulations, ABM establishes theoretical validity to explore potential impacts on social phenomena [61].

Current research applies ABM for group-scale studies at the building level. Yu et al. modeled office occupancy and interaction patterns as agents, creating five agent models based on organizational structures and survey data. By simulating different layouts of work and rest areas, the study measured the impact of environmental layouts on occupancy behavior [59]. Cheliotis et al. focused on four typical visitor types as agents, establishing a system in an open park. They validated computer-simulated behavioral patterns against real-world behaviors and explored the influence of open public spaces on group behavior [62]. Uddin proposed a hybrid modeling approach combining ABM, system dynamics (SD), and building information modeling (BIM) to predict the energy-saving effects of residents' behavior under indoor layout configurations. The study enhanced model reliability, credibility, and robustness by validating with a real data collection system using custom sensors [63].

Garcia et al. developed a dynamic model through ABM, simulating adult psychological and environmental factors to aid in exploring the formation and evolution of leisure-time PA group patterns [64]. Frerichs et al. engaged adolescents in the construction and simulation of ABM, investigating factors influencing PA time and confirming ABM's ability to visualize challenging-to-represent PA activities [65]. In the study of PA in children, Almagor et al. explored children's PA as a complex system using ABM, focusing on 9–11-year-old urban children. The research suggested that interventions in outdoor games, school sports, and active travel effectively increased children's PA time [66]. In the prevention and treatment of childhood obesity, Zhang's and Hennessy's teams utilized ABM to simulate interventions targeting children and communities to explore ways to increase children's PA time, consequently reducing the obesity rate among school-aged children [67,68]. Yang and Diez-Roux utilized ABM to simulate American children's neighborhood environments, revealing that increased school size and population density significantly enhance the likelihood of active travel to school [69]. Zhang employed social network analysis and ABM, using explicit micro-level rules to predict future behaviors, to assess whether network interventions could effectively enhance children's PA levels [70].

While ABM has demonstrated extensive applications across diverse fields, current research tends to focus predominantly on specific aspects such as building occupant behavior, social dynamics, and PA patterns in children. However, there remains a gap in the construction of school environment models, notably neglecting the impact of school spatial layouts on agent behavior simulation. Moreover, there is a lack of research integrating ABM with children's PA while presenting the results at a spatial level.

2. Methods

This study utilized an ABM approach to simulate and analyze students' PA during recess. First, three Chinese primary schools with representative spatial layouts were selected, and data were collected on school area, activity space types, layout features, and student attributes, including age, gender, stride length, recess activity types, locations, and clustering preferences. Then, the data were processed to examine correlations between recess activities and spatial features through data fitting. Based on these findings, an ABM was developed in AnyLogic software, incorporating clustering social force preferences, to simulate PA under different campus layouts. Finally, the model's validity was tested against field data, and the impact of activity space layouts on students' PA levels was analyzed.

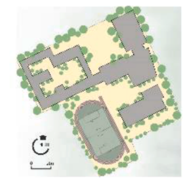
2.1. Study area

This study screened several primary schools in central and eastern China, focusing on campus environment, site area, diversity of activity spaces, and the proximity of activity areas to the buildings. Three representative schools were selected: NY Primary School in Hefei, Anhui Province; SY Primary School in Huai'an, Jiangsu Province; and SY Primary School in Wuchang, Hubei Province, as the primary subjects for research and analysis. All three schools are situated in high-density old urban areas, covering land areas ranging from 26,000 to 38,900 m^2, minimizing the influence of size-related variables. The architectural layouts of these schools reflect common spatial designs in Chinese primary schools, including ’E, ’ ’U, ’ ’L’ and line shapes. Other schools were excluded due to limited activity space types or unrepresentative layouts. All three selected schools provide necessary outdoor activity spaces, with each featuring at least three distinct types of activity areas. However, subtle differences exist in aspects such as campus environment, the richness of activity space types, and the distance from activity areas to teaching buildings among these selected schools. This study aims to analyze and compare these spatial distinctions to explore how school layouts and activity space configurations influence students' PA levels. These school activity space environment characteristics are presented in Table 1.

2.2. Data source

Data were collected through questionnaires and field observations. Questionnaires covered aspects such as PA volume, type, group size, and location. Fieldwork involved full-day observations at each school, monitoring six 10-minute recess periods (three in the morning and three in the afternoon). The dataset provided cross-sectional insights into students' choices of activity spaces, PA types, and group sizes across

Table 1
Basic environmental information of the sample primary schools.

Primary school	Site area	Building area	Layers	Layout mode	Activity space			The site plan
					Type	Area (m^2)	Shortest distance from the school building(m)	
NY Primary School in Hefei	26,700 m^2	7900 m^2	3 floors	E-shape	Entrance square	1329	43	
					Playground	3104	45	
					Courtyard	784	17	
SY Primary School in Huai'an	26,000 m^2	35,000 m^2	4~6 floors	U-shape;line shape	Entrance square	1850	17	
					Playground	3000	12	
					Courtyard	1395	8	
SY Primary School in Wuchang	38,839 m^2	24,575 m^2	6 floors	L-shape; line shape	Entrance square	1550	26	
					Playground	3256	9	
					Courtyard	1083	6	
					Outdoor platform	570		

different grade levels during recess [71–73]. We conducted a descriptive statistical analysis on the PA data of students from three sampled primary schools, 801 valid data points, including 406 boys (51 %) and 395 girls (49 %), with a balanced gender distribution. The sample was further divided into three grade-level groups based on cognitive and physical developmental differences, which may influence recess activity needs and behaviors [74]: junior grades (239 students), middle grades (219 students), and senior grades (343 students), with a proportional distribution of approximately 3:3:4.

Students engage in various types of activities within outdoor spaces on the school premises, including necessary behaviors, spontaneous behaviors, and social behaviors. The intensity of PA is conveniently and practically represented by the Metabolic Equivalent of Task (MET_y), where higher levels of PA correspond to higher MET_y values, and vice versa. The categorization of students' PA into four intensity levels is based on the resting metabolic rate as the baseline MET_y value: sedentary PA (SED) ($\leq$ 1.5 METs), low-intensity PA (LPA) ($\geq$ 1.5 and < 3 METs), moderate-intensity PA (MPA) ($\geq$ 3 and < 6 METs), and vigorous-intensity PA (VPA) ($\geq$ 6 METs) [75]. In this study, the intensity of PA was quantified using estimated kilocalories. The calculation was based on the following formula:

$$BMR_{avg}(kcal/minutes) = [17.686 \times Weight(Kg) + 658.2]/1440 \quad (1)$$

$$EstimatedKilocalories = MET_y \times BMR_{avg} \times Minutes \quad (2)$$

Here, the MET_y value represents the metabolic equivalent for a specific physical activity. BMR_{avg} typically refers to the average Basal Metabolic Rate for a specific group or population. By multiplying MET_y with BMR_{avg} and the duration of the activity, the estimated kilocalories expenditure for the activity can be calculated. To account for the variations in body weight across different age, average body weight was used to calculate BMR_{avg} in this study. Table 2 provides an overview of the MET_y linked to necessary, spontaneous, and social behaviors [76, 77]. Table 3 summarizes the average body weight and BMR_{avg} values of primary school students across different grade-level groups. Tables 4 and 5 analyze the students' PA in recess from the aspects of the PA types, and PA spatial location, respectively.

Table 2
Metabolic equivalent of different activities (MET_y).

Activity nature	Forms of activity	Activity intensity	MET_y value
Necessary behavior	Going to the bathroom	LPA	2.5
	Sleeping	SED	1.2
Spontaneous behavior	Learning (Sitting)	SED	1.4
	Watching	SED	1.4
Social behavior	Chatting (Standing)	LPA	1.8
	Walking	LPA	2.9
	Playing	MPA	5.8
	Sporting	VPA	7.0

Table 3
Metabolic equivalent of different activities (MET_y).

Grades	Age	Standard weight for boys(kg)	Standard weight for girls(kg)	Average weight(kg)	BMR_{avg}
Junior	7	26.6	24.7	25.65	0.79
	8	29.9	27.6	28.75	
Middle	9	33.6	31.3	32.45	0.88
	10	37.2	35.5	36.35	
Senior	11	41.9	40.6	41.25	0.99
	12	46.6	44.5	45.55	

2.3. Model construction

Based on the social force model, the AnyLogic platform's ABM system can more accurately describe agent's movement patterns. AnyLogic software enables the construction of simulation objects in physical spaces, such as architectural components like walls, pillars, elevators, and service facilities [78]. By defining the starting and destination points of crowds based on research requirements, agent behavior processes are established, allowing for the visualization of data such as a

Table 4
The distribution of the type of students' PA of different grades in recess.

School	Grade	Learning / reading	Watching	Chatting	Walking	Chase/ playing	Sporting	Total
NY Primary School in Hefei	Junior	2	26	24	13	15	23	103
	Middle	4	21	20	13	7	31	96
	Senior	9	17	22	14	5	34	101
SY Primary School in Huai'an	Junior	6	21	18	5	10	16	76
	Middle	8	20	17	6	6	15	74
	Senior	11	18	15	8		18	76
SY Primary School in Wucahng	Junior	2	16	14	4	9	15	60
	Middle	3	7	10	7	8	14	49
	Senior	17	31	38	26	10	44	166
Total average probability	Junior	5.02 %	23.01 %	25.10 %	11.72 %	15.48 %	19.67 %	
	Middle	8.22 %	22.37 %	22.37 %	11.87 %	9.59 %	25.57 %	
	Senior	11.95 %	16.62 %	24.20 %	14.29 %	6.71 %	26.24 %	
Total	Junior	12	55	60	28	37	47	239
	Middle	18	49	49	26	21	56	219
	Senior	41	57	83	49	23	90	343

Table 5
The distribution of the spatial location of students' PA of different grades in recess.

School	Grade	Corridor	Semi-indoor/ outdoor platforms	Courtyard	Entrance square	Playground	Total
NY Primary School in Hefei	Junior	27	41	14	8	13	103
	Middle	24	36	11	11	14	96
	Senior	26	34	15	10	16	101
SY Primary School in Huai'an	Junior	23	35	7	3	8	76
	Middle	19	32	8	6	9	74
	Senior	21	29	10	5	11	76
SY Primary School in Wuchang	Junior	25	21	7	2	5	60
	Middle	18	18	4	1	8	49
	Senior	53	43	21	5	44	166
Total average probability	Junior	31.38 %	40.59 %	11.72 %	5.44 %	10.88 %	
	Middle	27.85 %	39.27 %	10.50 %	8.22 %	14.16 %	
	Senior	29.15 %	30.90 %	13.41 %	5.83 %	20.70 %	
Total	Junior	75	97	28	13	26	239
	Middle	61	86	23	18	31	219
	Senior	100	106	46	20	71	343

density, quantity, and walking time – which serve as evaluation metrics.

In this study, an ABM simulation model is constructed using the AnyLogic platform. Through experiments, simulated data, including agent density in various activity spaces and the duration of pedestrians' stay in specific areas, are obtained to validate the actual usage of activity spaces and characterize the PA of primary school students in these spaces. The overall technical workflow of the study, as depicted in Fig. 1, encompasses the data collection stage, model configuration stage, and simulation stage. During the model configuration stage, it encompasses two main components: physical environment modeling and agent behavior modeling.

2.3.1. Environment

Spatial environment modeling involves the description and construction of the infrastructure for simulating the current state of primary school campuses. The three groups of sample models were divided into different functional types of areas, including the main school building, office buildings, corridors, courtyards, squares, playgrounds, etc., and the entrances and exits of each school building were defined (Fig. 2). Each type exhibits different characteristics: the teaching building is a place for students to engage in theoretical studies, the corridor is a passageway for students to reach the activity areas, platforms are larger staging spaces within the corridor, and outdoor spaces such as courtyards, squares, and playgrounds are a place for PA during recess. Due to factors such as weather, time, or personal preference for lower-intensity PA, some Chinese students choose to engage in PA in corridors and on platforms. As a result, these areas function not only as circulation spaces but also as venues for recess activities. In subsequent simulations, activities in corridors and on platforms were categorized into two types: transit and stationary.

The specific construction steps are as follows: Firstly, distill satellite maps and typical spatial layout CAD drawings for three primary school campuses, create a comprehensive color site plan, and mark important entrance spaces, such as school entrances and teaching building entrances. Secondly, import the exported JPG-format color site plan into AnyLogic software at specified proportions. Within the AnyLogic working window, utilize the 'Wall' tool in the 'Pedestrian Library' to draw buildings, roads, traffic spaces, simulation boundaries, etc., completing the drawing of the spatial environment base map. The final 3D model of the environment is depicted in Fig. 3.

2.3.2. Agent

In the ABM simulation system for PA among primary school students, unlike conventional Agent parameter settings, adjustments and configurations are required specifically for the characteristics of primary school student agents. The parameter settings primarily involve individual features and social force characteristics. This study's parameter configuration is based on analysis of students' PA data in Section 2.2. Moreover, agents' PA and participation in certain activities are influenced by peer behavior. Children are more likely to engage in activities when others are present. To capture these influences, social relationships are created among agents to represent friendship networks. While participating in activities together, agents adjust their preferences based on the inclinations of their friends. Table 6 compiles the group scales of peer clusters during physical activities in the school recess. These findings serve as a basis for configuring various indicators related to children's agent behavior.

Agent Characteristics: The gender ratio is 1:1, ages range from 6 to 12 years old, and grades are divided into junior, middle, and senior grades (with a ratio of 1:1:1). The average walking speed is 1.1 m/s

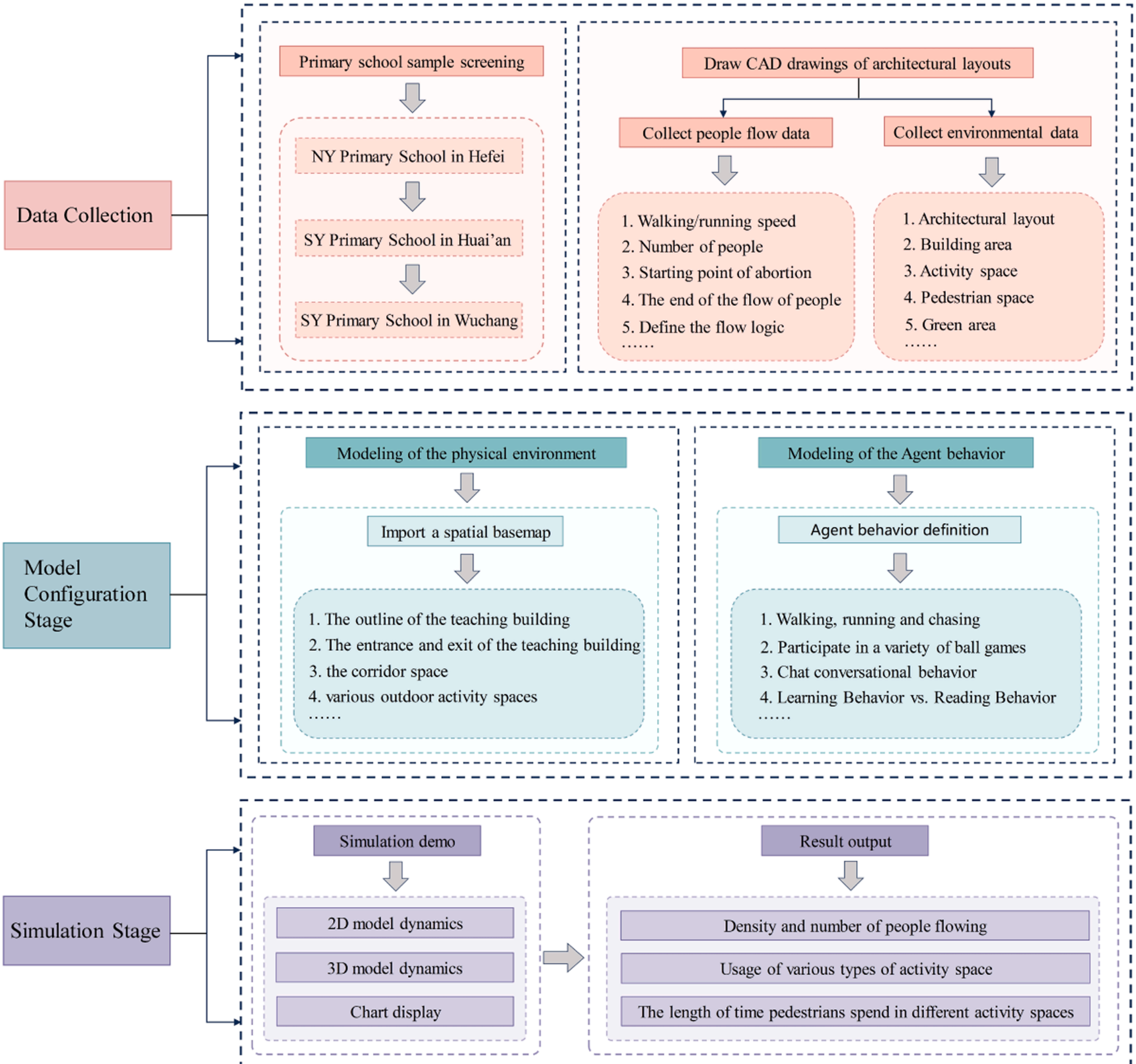


Fig. 1. The overall technical flow chart of the study.

(with walking speeds ranging from 0.8 to 1.1 m/s and running speeds from 1.1 to 1.4 m/s). Personality traits are distributed from introverted to extroverted.

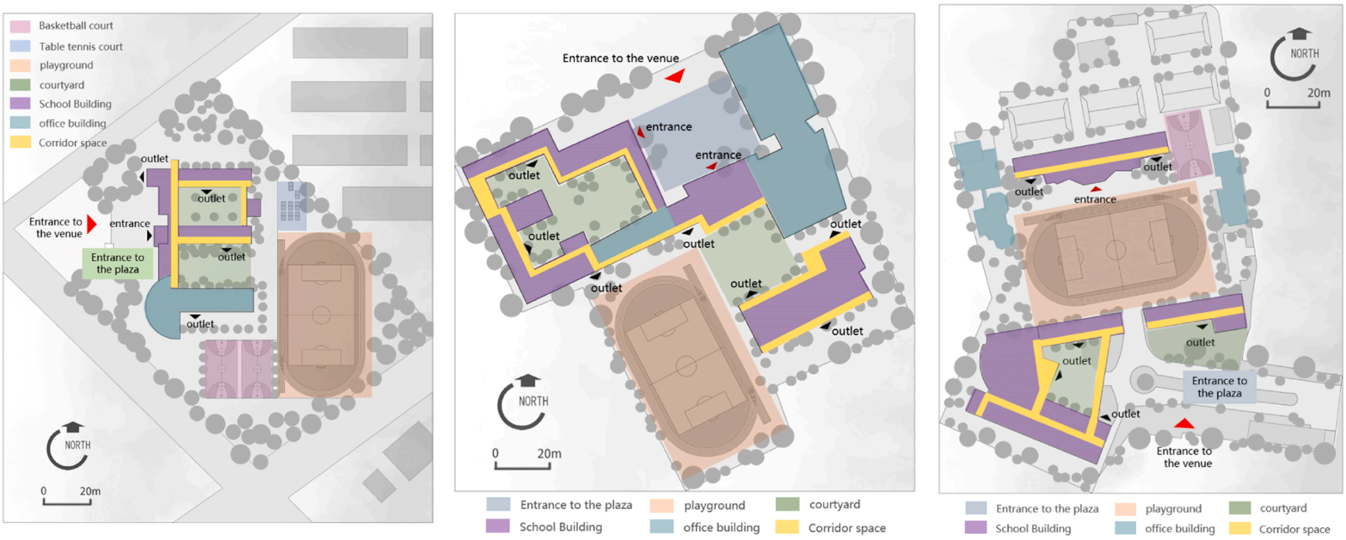
Agent Social Network: In PA, there can be both individual and group activities. In individual activities, observational behavior may influence behavioral choices. In group activities, the presence of friends influences an agent's decisions, with each additional participant slightly increasing the likelihood of others joining the same activity. This social reinforcement is modeled by assigning incremental weights to activity choices, capturing the effects of both individual observation and group influence.

2.3.3. Process overview and scheduling

According to the schedule of Chinese primary schools, the available times for PA for primary school students during school hours mainly include short recesses, long recesses, lunchtime, and organized school sports activities. This project focuses on studying primary school students' PA activities during recess periods. The simplified daily schedule for students is shown in Fig. 4, and the simulation of children's activities during recess times is carried out. The activities of primary school students during recess can be broadly categorized into essential behaviors, spontaneous behaviors, and social behaviors. These behaviors have different priorities and are also influenced by the complex factors of the school environment and surrounding individuals. In the Agent simulation process, the logic behind children's choices during recess activities is defined, as illustrated in Fig. 5.

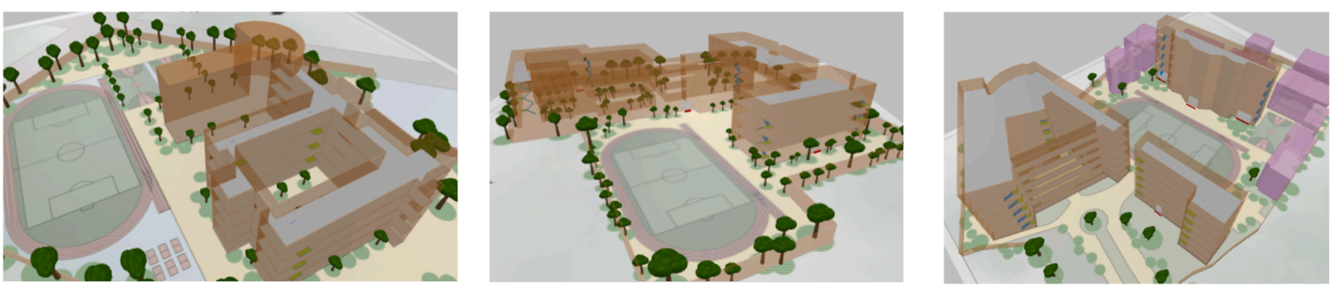
2.3.4. Controller decision-making logic

The decision-making logic for children Agents engaging in recess activities is complex, primarily influenced by individual characteristics, social forces, spatial environment layout, and temporal factors. This project primarily investigates the impact of school spatial layout on primary school students' choices of recess activities. It employs multiple linear regression analysis to establish mathematical relationships, and the feasibility of the model is validated through correlation analysis.



(a) (b) (c)

Fig. 2. (a) The site plan of NY Primary School; (b) The site plan of SY Primary School in Huai'an; (c) The site plan of SY Primary School in Wuchang.



(a) (b) (c)

Fig. 3. (a) NY Primary School 3D model; (b) SY Primary School in Huai'an 3D model; (c) SY Primary School in Wuchang 3D model.

Table 6
The distribution of the social scale of students' activities of different grades in recess.

School	Grade	Solo activity	Groups of 2∼3 people	Groups of 4∼5 people	Groups of 6+ people	Total
NY Primary School in Hefei	Junior	17	40	28	18	103
	Middle	15	35	26	20	96
	Senior	15	34	33	19	101
SY Primary School in Huai'an	Junior	11	32	21	12	76
	Middle	10	31	20	13	74
	Senio	9	29	24	14	76
SY Primary School in Wuchang	Junior	8	36	10	6	60
	Middle	8	16	14	11	49
	Senior	5	48	91	27	166
Total average probability	Junior	15.06 %	45.19 %	24.69 %	15.06 %	
	Middle	15.07 %	37.44 %	27.40 %	20.09 %	
	Senio	8.45 %	32.36 %	41.11 %	17.49 %	
Total	Junior	36	108	59	36	239
	Middle	33	82	60	44	219
	Senior	29	111	141	60	343

Finally, individual characteristics and social forces are incorporated into the AnyLogic modeling. The ABM simulation was conducted exclusively on the ground floor of the building and the site plan. The corridors and ground-floor spaces serve as the necessary pathways for students to access outdoor activity spaces. To accurately capture PA within corridors and platforms, only stationary activities conducted in these spaces are recorded, excluding transit-related movements. The aim of the simulation is to explore the different choices and usage patterns of students as they move from classrooms, their starting point, to outdoor activity spaces.

In this project, the probability of children choosing specific recess activities is influenced by multiple factors. The optimal combination of several independent variables is collectively used to predict or estimate the dependent variable. Therefore, a multiple linear regression model is employed for simulation. Let Pa,i(t) represent the probability of Agent *i* engaging in activity '*a*' during recess time '*t*':

$$P_{a,i(t)} = F_{a,i} \cdot E_a \cdot S_{a,i} \cdot T_t \tag{3}$$

Here, $F_{a,i}$ is Agent *i*'s preference for recess activity, E_a represents the influence of the campus spatial environment on activity '*a*', $S_{a,i}$ signifies the impact of the surrounding social environment on Agent *i*, and T_t denotes the temporal influence during recess activities..

For the main research question— the impact of campus spatial layout on primary school students' choices of recess activities— the participation proportions in various activities for each school in the research sample were statistically analyzed. The campus activity space area (S_e) and distance (D_e) were transformed into spatial attractiveness (A_e) using the formula:

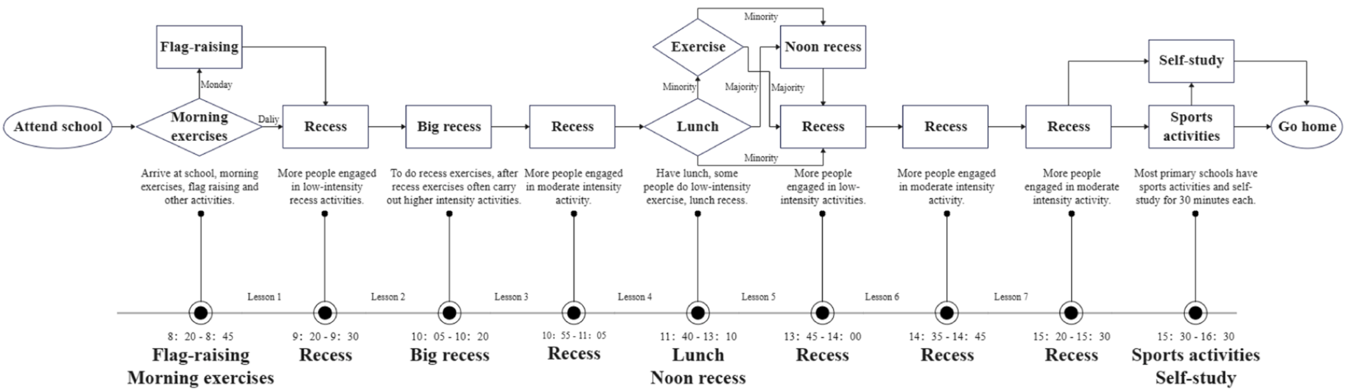


Fig. 4. The schedule of the students during their school time.

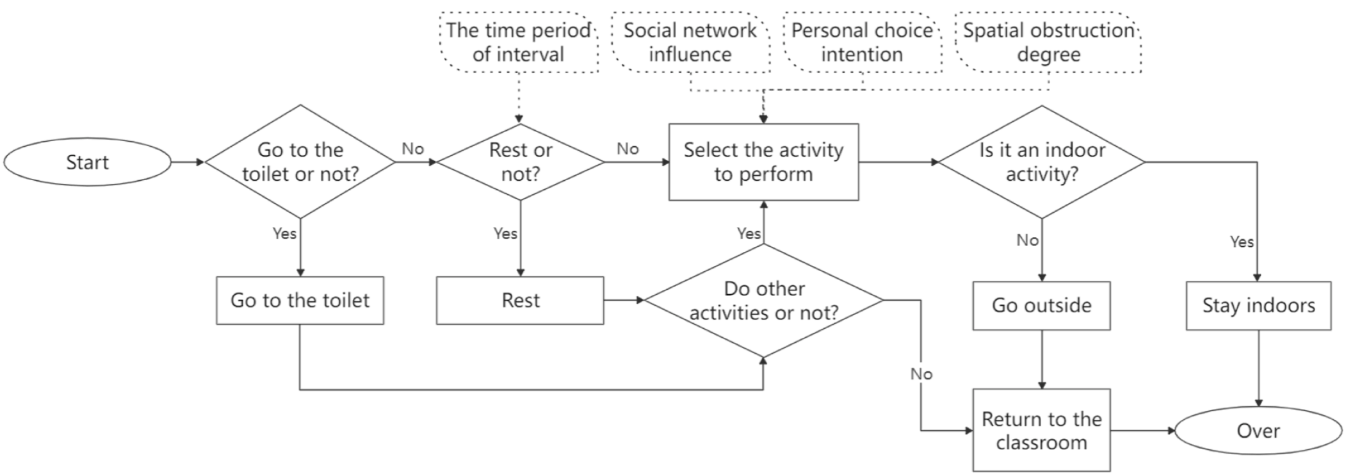


Fig. 5. Agent behaviour flowchart.

$$A_e = \frac{S_e}{S_{\text{total}}} - \frac{\sqrt{D_e}}{100} \tag{4}$$

The final statistical results are presented in Fig. 6. Taking the attractiveness of each campus activity space as the independent variable and the probability of students choosing various activities as the dependent variable, a multiple linear regression analysis was conducted, and the results are shown in Table 7. When the adjusted R^2 is closer to 1, the influence of the independent variable on the dependent variable is

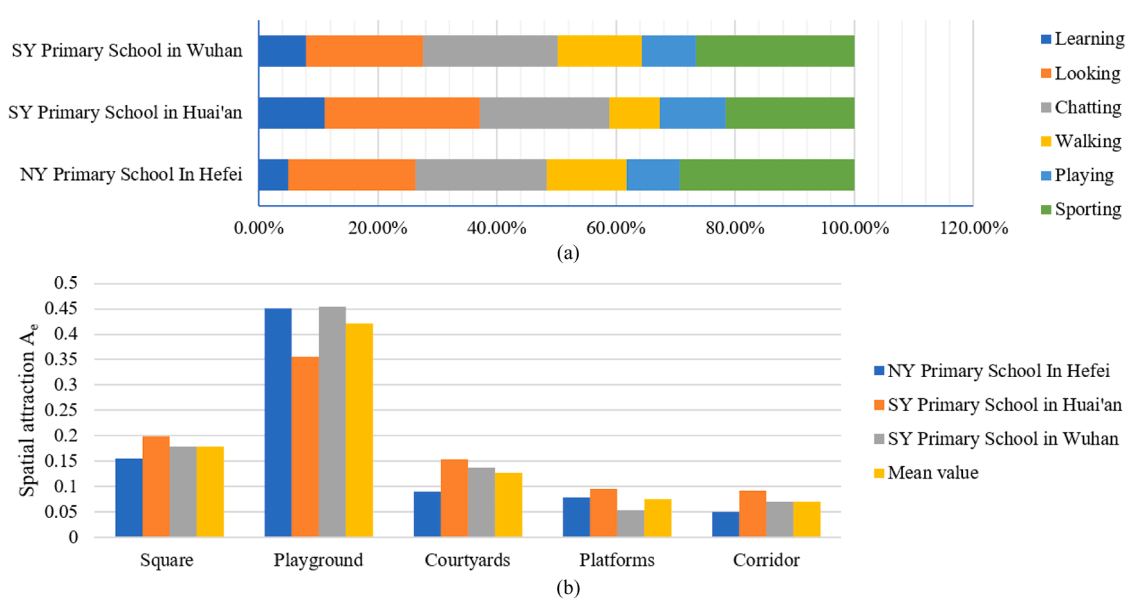


Fig. 6. (a) The proportion of school recess PA choice; (b) Corresponding Spatial attraction A_e for each site.

Table 7
The results of multiple linear regression analysis were analyzed by SPSS software.

Activity	The adjusted (R^2)	Spatial influencing factor	Significance (P)	Unstandardized coefficient (B)	Constant (B)	VIF
Looking on	0.995	Semi-indoor/ outdoor platforms	0.043	1.198	0.084	1.207
		Corridor	0.037	0.670	0	1.207
		Courtyards	0.126	0	0	10.519
		Entrance Square	0.409	0	0	5.186
Chatting	0.996	Semi-indoor/ outdoor platforms	0.006	−0.123	0.225	1.019
		Courtyards	0.009	0.051	0	1.019
		Corridor	0.156	0	0	6.428
		Entrance square	0.307	0	0	32.283
Walking	0.993	Corridor	0.039	−4.205	0.17	1.222
		Courtyards	0.048	1.971	0	1.222
		Entrance square	0.163	0	0	6.049
		Semi-indoor/ outdoor platforms	0.308	0	0	5.091
Chase/ playing	0.718	Playground	0.029	−0.176	0.105	1.240
		Semi-indoor/ outdoor platforms	0.043	0.151	0	1.240
		Entrance square	0.023	0.307	0	1.240
		Courtyards	0.308	0	0	8.358
Sporting	0.990	Playground	0.043	1.019	−0.262	3.031
		Semi-indoor/ outdoor platforms	0.049	1.232	0	3.031
		Courtyards	0.193	0	0	5.186
		Entrance square	0.353	0	0	5.823

greater. Significance ($P < 0.05$) indicates a significant impact of the independent variable on the dependent variable, and when the VIF value is <5, it proves that there is no collinearity between the independent variables.

After filtering the results of the regression analysis, the final expression of the impact of spatial environment layout on different activities is as follows:

$$E_{a(\text{activity})} = C_{0a} + C_{1a} \cdot A_{\text{rooftop}} + C_{2a} \cdot A_{\text{corridor}} + C_{3a} \cdot A_{\text{courtyard}} + C_{4a} \cdot A_{\text{entranceplaza}} + C_{5a} \cdot A_{\text{playground}} \tag{5}$$

where ‘activity’ includes ‘looking on’, ‘chatting’, ‘walking’, ‘chase/ playing’, ‘sporting’ and ‘learning/reading’. Here, C_{0a} represents a constant, and C_{1a}, C_{2a}, C_{3a}, C_{4a}, C_{5a} are coefficients indicating the attractiveness of different activity types to the respective school campus spaces.

Let ‘learning/reading’ represent the probability of Agent i choosing to learn or read in the classroom during recess under the influence of campus activity space layout ‘a’. As compared to other activities, it is less influenced by the spatial environment, hence:

$$E_{a(\text{learning/reading})} = 1 - E_{a(\text{lookingon})} - E_{a(\text{chatting})} - E_{a(\text{walking})} - E_{a(\text{chase/playing})} - E_{a(\text{sporting})} \tag{6}$$

3. Results

3.1. PA duration time analysis

The distribution of average activity duration time of 2400 agents in 10- minute recess, generated by 3 simulations of the different activity space is presented in Fig. 7a. At NY Primary School, the platform witnesses the longest duration of activity engagement among agents. Conversely, in the other two schools, the corridor emerges as the space

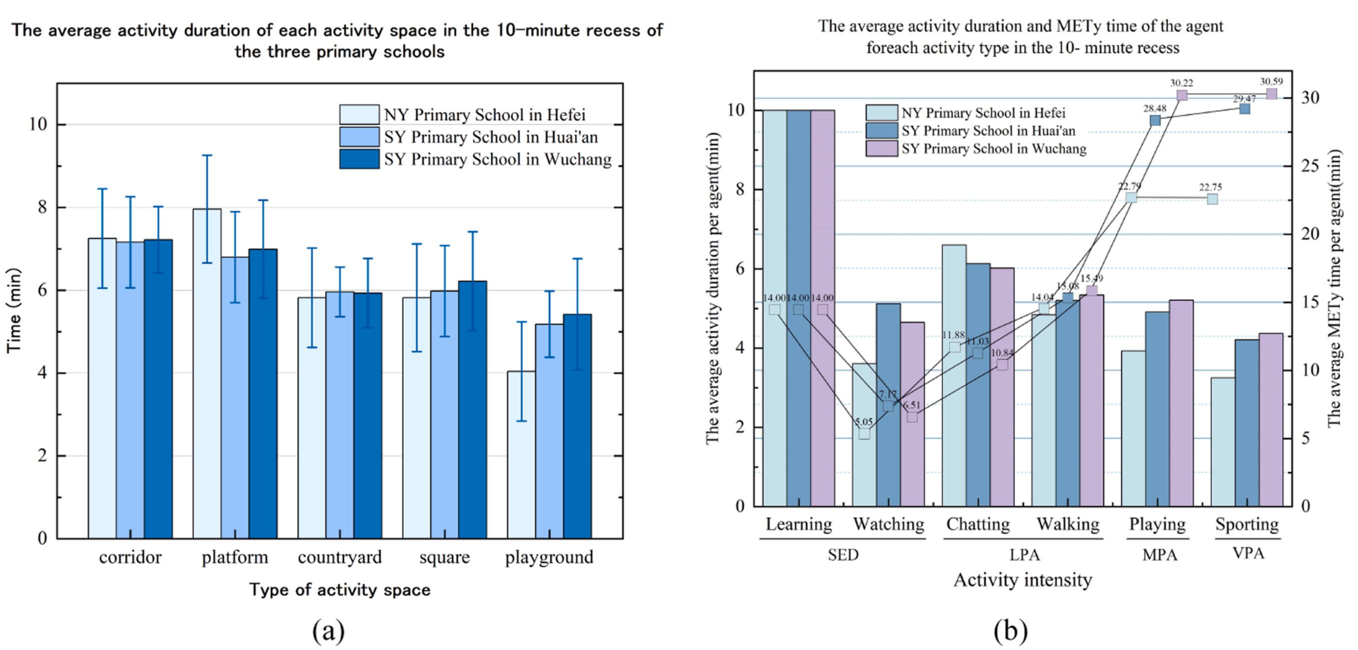


Fig. 7. (a)The average activity duration of each space in the 10-minute recess of the three primary schools; (b) The average activity duration and MET_y time of the agent for each activity type in the 10-minute recess.

where agents spend the most time during recess, closely followed by the platform. This phenomenon may be attributed to the proximity of both the corridor and platform to the classrooms. Most outdoor or semi-indoor platforms are located within the school buildings, sometimes even on the same floor as the classrooms. Consequently, spaces in proximity are prioritized by agents. Across all three schools, the playground records the least activity time. In NY Primary School, the average agent's activity time in the playground was only 50 % of the average activity time in the corridor.

Fig. 7b presents the average PA time/10-minute recess for each type of the PA or PA levels in the three primary schools. In the three schools, learning is the activity with the longest average duration during recess, at 10 mins. This is partly due to recess time being taken up by extra teaching or written exercises, as high academic pressure in China often pushes schools to maximize learning time to improve student performance [79]. The shortest activity is sporting, lasting between 3 and 4.5 mins, as VPA, which is less than half of the 10-minute duration. Regarding MPV playing, except for SY Primary School in Wuchang where it exceeds half the time at 5.21 mins, the other two schools are below half the time. By multiplying the MET_y value by the duration of activity in minutes, it is possible to calculate a standardized metric that reflects the time spent on activities of a specific intensity (e.g., MET-minutes or MET-hours). The longer the MET_y time, the greater the potential health benefits [80]. This metric can be used to assess the impact of the spatial layout on the students' physical health. Despite the shorter duration of MVPA activities, their MET_y time is much higher than that of SED, with sporting having the highest MET_y time. However, there is still some distance from the recommended minimum of 60 mins of MVPA per day, highlighting the necessity of promoting MVPA during recess.

Fig. 8 presents the total kilocalories consumed by agents of different grades during each PA type in the three primary schools. By multiplying the MET-hours by students BMR in kilograms per hour one can estimate the kilocalories energy expenditure of an activity [76]. In SED, such as learning, junior-grade students exhibited the highest energy expenditure, while in watching activities, middle-grade students expended the most. For LPA, such as chatting, the caloric expenditure was relatively evenly distributed across junior, middle, and senior grades. MPA showed greater energy expenditure among junior-grade students in two schools, while senior-grade students expended less. Notably, in VPA, senior-grade students in all three schools expended more than double the kilocalories of junior-grade students.

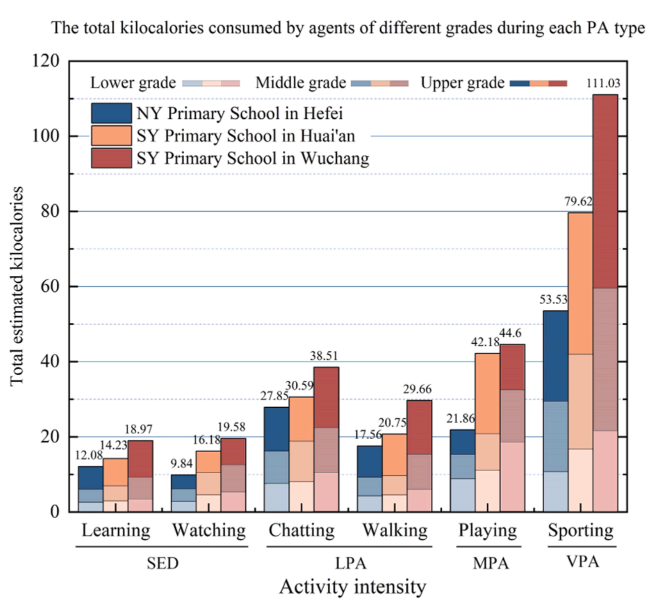


Fig. 8. The total kilocalories consumed by agents of different grades during each PA type.

3.2. PA occupancy density distribution

The visual output of the simulation showed the locations of the agents separated by the different actions throughout the simulation. The visual output was displayed heat maps showing the locations of the different agents and occupancy density distribution that occurred during the simulation (Fig. 9).

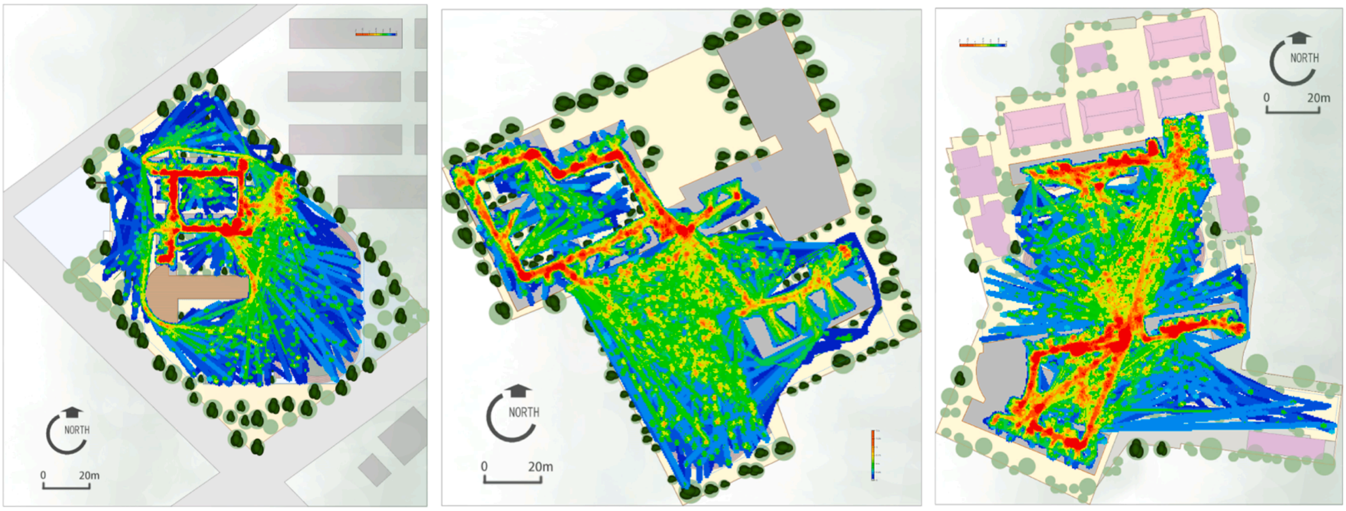
Comparisons among the three layouts reveal that corridors are the most popular stationary spaces for primary school students during recess. Corridors are closest to the classrooms, making them the shortest route preferred by most students. In contrast to the other two layouts, the E-shaped layout exhibits the lowest activity intensity in both playgrounds and courtyards. Particularly, the playground's activity intensity is depicted in deep blue, indicating the fewest number of students choosing to engage in activities there during recess. The playgrounds in the l-shaped and linear layout combinations exhibit the highest activity intensity. This can be explained by the fact that in the l-shaped layout, the playground is situated near the longer side of the classroom building, and there is a greater contact area between the building and the playground, resulting in higher accessibility for students. Conversely, in the E-shaped layout, the short side of the classroom building is adjacent to the playground, with minimal contact between the building and the playground, leading to more turns in the path and lower accessibility for students. Comparatively, the courtyard in the U-shaped layout demonstrates higher activity intensity, possibly due to the enclosed space being more conducive to children's engagement, while the central outdoor courtyard offers students more opportunities for interaction.

To validate the impact between different PA intensities and activity space layouts, each PA intensity was tested individually. Three simulation runs were executed for each parameter set. Fig. 10 displays the spatial footprints of four different PA intensities and the density distribution of agents in activity spaces. SED including learning and watching, shows the highest density within the classroom buildings. Specifically, the probability of agents engaging in SED within the E-shaped layout is higher than the other two types. LPA including chatting and walking within the classroom buildings and courtyards, exhibits higher probabilities, with similar distribution probabilities across the three layouts. The pathways and greenery in the courtyards may influence the choice of walking in courtyards.

MPA (playing) shows a higher probability of occurrence in the enclosed courtyards surrounded by three sides of classroom buildings in the l-shaped layout, inside the classroom buildings in the E-shaped layout, and in the courtyards and entrance squares in the l-shaped layout. The distribution probability of MPA in the l-shaped layout is higher than the other two layouts, possibly owing to its connectivity between entrance squares, courtyards, and playgrounds, offering higher openness and accessibility. VPA (sporting) has the highest distribution probability in the playgrounds, but in the E-shaped layout, only a small portion of the playgrounds adjacent to the classroom buildings shows a higher probability. The highest probability of VPA occurs in the playgrounds of the l-shaped layout, especially in the areas close to the classroom buildings and along the pathways from the courtyards to the playgrounds. A potential explanation is the increasing number of agents engaging in sporting on the playground, triggering positive feedback that further encourages other agents in the social network to participate in outdoor sports, thereby promoting additional accumulation of daily MVPA [81].

3.3. Verification test

In the ABM, we define behaviors for individual agents and assign values to multiple parameters governing their decisions and actions in response to diverse social and environmental conditions. The simulation



(a) (b) (c)

Fig. 9. (a) Heat maps of NY Primary School; (b) Heat maps of SY Primary School in Huai'an; (c) Heat maps of SY Primary School in Wuchang.

execution reveals the complex interplay among these parameters, agent interactions, and environmental dynamics, resulting in emergent effects within the PA system. These emergent effects cannot be directly derived from the individual-level behaviors of agents but arise from the holistic interactions encapsulated within the simulation. The correlation between the characteristics of the agents in the different scenarios and the PA results indicates the feasibility of the simulation results.

The purpose of this paper is to demonstrate the construction mechanism of the model, particularly focusing on the PA model tailored for primary school students. Although the validation of the model extends beyond the scope of this study, some observational evidence (Section 2.2) was utilized to identify similarities between occupancy patterns in real-life and simulated results in Section 3.2. We conducted three simulation experiments, each with the same parameter settings, and recorded the proportion of agents choosing each type of PA in each experiment (Fig. 11). These data were employed to ascertain whether agents tended to occupy locations and activities similar to students in actual survey data. The model appears to present clustering effects similar to survey studies, especially when comparing the observed proportions of various PA to the simulated proportions during recess: the first experiment showed the largest discrepancy with real data, while in the second experiment, the data for SY Primary School in Huai'an closely resembled the survey data, with walking (8.71 %) having the fewest participants, differing from the actual (8.41 %) by only 0.3 %. In the third experiment, the overall experimental data closely matched the survey data, particularly for SY Primary School in Wuchang, where sporting activities (26.46 %) accounted for the highest proportion, differing from the actual (26.55 %) by less than 0.1 %. This indicates that with increasing trial iterations, the simulated results converge towards the real data.

4. Discussion

4.1. The impact of school activity space layout on PA

This study introduces an ABM to simulate and analyze the relationship between the layout of school activity spaces and children's PA during recess. The simulation results indicate that spatial layouts significantly influence children's PA levels, with proximity and accessibility to activity areas like playgrounds and corridors playing a crucial role in determining the intensity and duration of PA. These findings align with previous studies that emphasize the importance of spatial configurations in promoting physical engagement during unstructured periods, such as recess [51,82]. The ABM effectively captures aggregate patterns of PA within primary school environments, demonstrating that corridors and platforms, being closer to classrooms, tend to facilitate more prolonged engagement in activities. Conversely, playgrounds, especially those with less accessible layouts, witness lower levels of PA. This finding is consistent with the work of Martensson et al., who noted that spaces close to buildings are often more utilized, offering ample opportunities for socialization and PA [83]. However, the lack of PA intensity in playgrounds across all three schools, particularly in E-shaped layouts, highlights the influence of spatial layout on activity choices. The playgrounds' relative inaccessibility, especially in the E-shaped layout with limited direct contact with classroom buildings, serves as a barrier to maximizing PA during recess. The findings also indicated that promoting recess PA is challenging, but placing features near classrooms or entrances could be an effective strategy, as supported by previous studies [44,84]. Additionally, l-shaped and linear layouts with interconnected or open designs between entrance plazas, courtyards, and playgrounds, were found to encourage LPA and MPA. Existing research also corroborates that open spaces tend to foster higher levels of PA [54]. Kilocalories expenditure analysis across grades revealed that older students exhibit significantly higher levels of PA compared to younger students. Previous studies have also shown that younger students engage less in organized sports activities, whereas sports and movement-based games become increasingly popular with age. Strategies to encourage younger students to participate in more PA during recess should be further explored [85,86].

In terms of PA levels, the simulations reveal that MVPA, particularly sporting activities, occurs at relatively low levels across all schools. This is concerning given the well-documented benefits of MVPA for children's physical and mental health. Although MVPA playing is higher in some schools, it remains below the daily recommended 60 mins of MVPA, which suggests that recess periods alone are insufficient to meet children's PA needs. The results align with the growing body of literature advocating for the redesign of school activity spaces to better promote PA, particularly by providing students with more diverse choices for activity areas [87].

4.2. Strengths and practical implications

Our findings have implications for the implementation of health-oriented school layout interventions. This study highlights the necessity of considering various types of activity spaces and prioritizing the proximity between teaching buildings and activity areas. The results

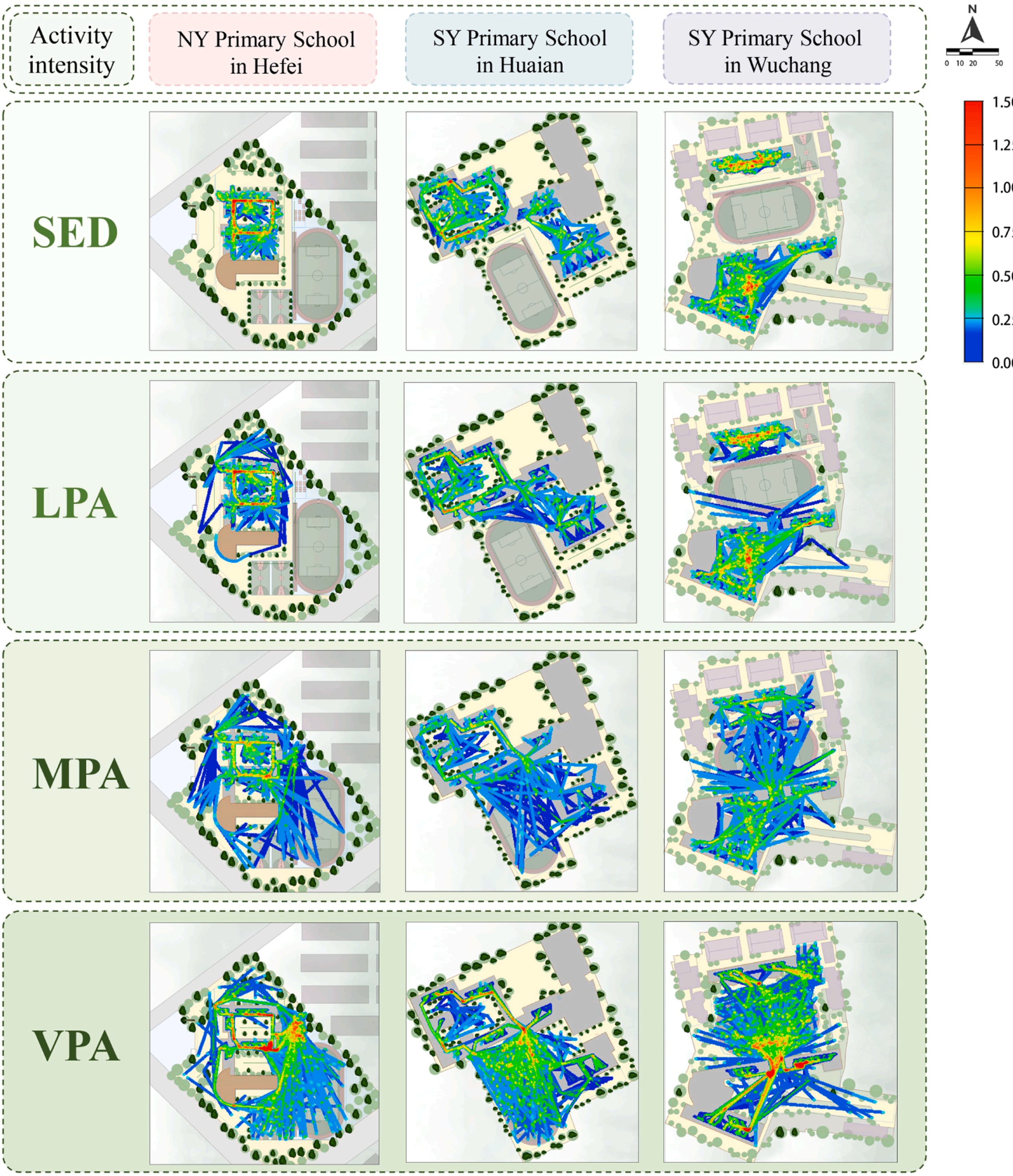
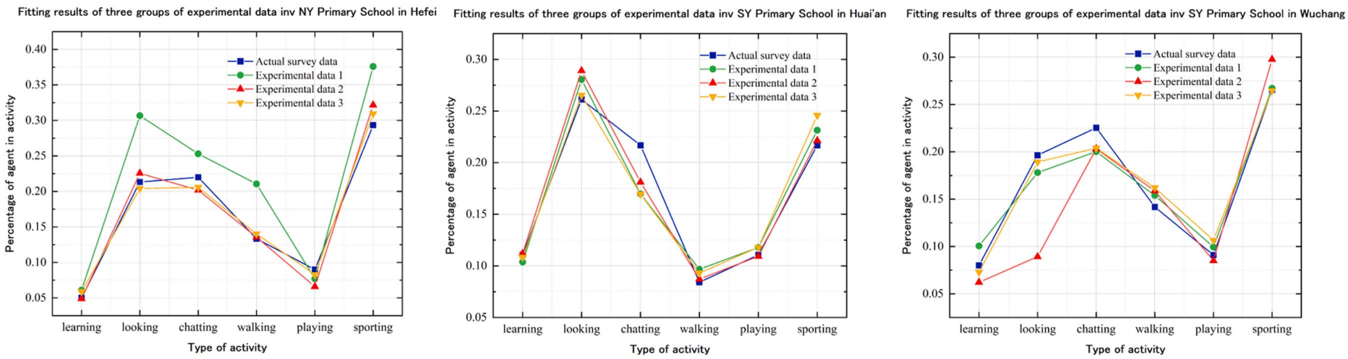


Fig. 10. Density distribution of activity spaces in three small schools for different PA intensities.

suggest that l-shaped and linear layouts, where the long side of the playground aligns closely with the long side of the teaching building, positively influence students' MVPA. Corridors, identified as spaces where students spend the most time during recess, currently facilitate primarily LPA. Therefore, it is recommended that corridors be redesigned with node-based interventions at high-traffic areas. Enlarging key corridor nodes or integrating them with platforms can provide greater space and opportunities for PA. These specific design recommendations expand existing primary and secondary school design standards, offering architects scientifically-informed strategies to enhance PA levels through spatial layout.

One of the strengths of this study lies in its use of ABM methods to assess the relationship between school space layout and PA. The ABM framework proposed in this study serves as a pre-assessment tool for



(a) (b) (c)

Fig. 11. (a) Fitting results of groups of experimental data in NY Primary School; (b) Fitting results of groups of experimental data in SY Primary School in Huai'an; (c) Fitting results of groups of experimental data in SY Primary School in Wuchang.

redesigning and optimizing school layouts to promote PA. Before implementing interventions, ABM simulations can be conducted based on the constructed Agent, process overview and scheduling, controller decision-making logic, and various environmental scenarios. These simulations predict PA outcomes under different layout modifications, providing quantitative and spatially visualized insights into whether these interventions will effectively enhance students' recess PA levels. By comparing PA outcomes before and after spatial interventions, this model offers a robust, data-driven framework for evaluating the effectiveness of proposed changes. Architects, planners, administrators, and policymakers can use these simulations to develop optimal design strategies that promote healthier, more active school environments. For local governments and decision-makers involved in promoting health-oriented campuses or renovating existing ones, PA should serve as a key evaluation metric [40]. The proposed ABM framework offers a measurable, visual tool to assess PA levels and durations, helping ensure design outcomes meet health goals.

The ABM approach offers a valuable means of examining emergent behaviors within complex systems, particularly in educational settings where multiple social and environmental factors interact. However, as with many ABMs, the model's ability to accurately replicate individual behaviors remains limited, as its primary focus is on reproducing aggregate patterns. This limitation is evident in the observed discrepancies between individual-level data and survey-based findings, especially in the first round of experiments. While subsequent simulations increasingly converged with real-world observations, the inherent variability in individual behaviors underscores the need for further refinement in ABM parameterization to improve predictive accuracy at finer levels of detail. Compared to traditional survey-based methods, ABM integrates spatial, social, and behavioral factors, replacing conventional equation models and micro-simulation techniques [88]. Its dynamic nature captures complex interactions between variables and enables real-time visualization of agent movements, making it less susceptible to the variability that often affects traditional methods [89].

4.3. Limitations and future work

Despite its contributions, this study is not without limitations. Firstly, while this study incorporated the social force model, considering factors like age, gender, and peer networks, and analyzed PA differences across age groups, the recess observations were limited. The sample size did not allow for generalizing the impact of social and organizational factors, such as school culture or recess management, on children's PA. Secondly, the study focused mainly on spatial layout factors, without considering other environmental elements like facility configuration, greenery, shading, or temperature, which also influence physical activity. While many studies have examined various factors affecting PA [90,91], this study concentrated on spatial layout and used the ABM model to visually represent its impact on PA levels. Thirdly, the small sample size limits the generalizability of the findings. Future research should include more schools from diverse socio-economic and geographic areas to validate and expand the results, and test the effectiveness of design interventions.

For future studies, it is recommended to enhance ABM models by including more social parameters and agent characteristics to better capture the complexity of student behavior. Integrating artificial intelligence algorithms into decision-making logic could allow for a more comprehensive consideration of factors affecting PA. Additionally, future work could attempt to integrate ABM with other modeling software to allow for more detailed environmental factor simulations. Finally, applying the ABM model across diverse regions and schools would strengthen its generalizability and empirical validation, thereby improving the robustness of the findings.

5. Conclusion

This study underscores the critical role of spatial layout in shaping children's PA behaviors during recess. By identifying key spatial factors, such as proximity and accessibility, that influence PA intensity, the findings contribute to the broader literature on school-based interventions for promoting children's physical health. Despite the benefits of MVPA, observed levels remain below the recommended 60 mins daily, highlighting the need for design interventions. The proposed ABM framework serves as a valuable tool for pre-assessing the potential impacts of design interventions on PA. By simulating agent behaviors and environmental interactions, it offers a data-driven approach for architects, planners, and policymakers to develop healthier school environments. Future research should build on this foundation by incorporating additional social and environmental factors into ABM models, broadening the scope of application to diverse regions, and empirically validating design strategies to ensure they effectively support students' physical and mental well-being.

CRediT authorship contribution statement

Jiameng Cui: Writing – review & editing, Writing – original draft, Methodology, Software, Validation, Visualization, Formal analysis, Data curation, Conceptualization. **Xue Meng:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Siming Qi:** Writing – original draft, Methodology, Software, Validation, Visualization, Formal analysis, Data curation. **Junwen Fan:** Writing – original draft, Methodology, Software, Validation, Visualization, Formal analysis, Data curation. **Wenxuan Yu:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Hongxing Liu:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xiang Wang:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Yu Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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